Prefix Sum

Definition: Used to perform multiple sum queries on subarrays efficiently. It involves precomputing an array where each element at index i stores the sum of all elements from the start up to i.

Sample Problem: Given an array nums, answer multiple queries about the sum of elements within a specific range [i, j].

Example: nums = [1, 2, 3, 4, 5, 6], range [1, 3]. Prefix Sum P = [1, 3, 6, 10, 15, 21]. Sum = P[3] - P[0] = 10 - 1 = 9.

Practice Questions:

Range Sum Query - Immutable (#303)

Subarray Sum Equals K (#560)

Two Pointers

Definition: Involves using two pointers that move through a data structure (usually a sorted array) toward each other or in the same direction to find a pair or subarray.

Sample Problem: Find two numbers in a sorted array that add up to a target value.

Example: nums = [1, 2, 3, 4, 6], target = 6. Pointers at 1 and 6 (sum 7, too high), move right pointer to 4. 1 + 4 = 5 (too low), move left pointer to 2. 2 + 4 = 6. Output: [2, 4].

Practice Questions:

Two Sum II (#167)

Valid Palindrome (#125)

3Sum (#15)

Sliding Window

Definition: Used to perform operations on a specific window size of an array or string. The window "slides" over the data to avoid redundant calculations.

Sample Problem: Find the maximum sum of a contiguous subarray of size k.

Example: nums = [2, 1, 5, 1, 3, 2], k = 3. First window [2,1,5] sum is 8. Slide to [1,5,1] sum 7. Slide to [5,1,3] sum 9. Max sum = 9.

Practice Questions:

Maximum Average Subarray I (#643)

Longest Substring Without Repeating Characters (#3)

Minimum Window Substring (#76)

Fast & Slow Pointers

Definition: Also known as Hare & Tortoise algorithm. Uses two pointers moving at different speeds to detect cycles or find the middle of a linked list.

Sample Problem: Detect if a linked list has a cycle.

Example: If a cycle exists, the fast pointer (2 steps) will eventually catch up to the slow pointer (1 step).

Practice Questions:

Linked List Cycle (#141)

Happy Number (#202)

Find the Duplicate Number (#287)

In-place Reversal of a Linked List

Definition: Reverses the nodes of a linked list without using extra memory by modifying the pointers.

Sample Problem: Reverse a linked list from position m to n.

Practice Questions:

- Reverse Linked List (#206)
- Reverse Linked List II (#92)
- Swap Nodes in Pairs (#24)

Monotonic Stack

Definition: A stack that maintains elements in a specific order (increasing or decreasing) to find the next greater or smaller element.

Sample Problem: Find the next greater element for each element in an array.

Example: nums = [2, 1, 2, 4, 3]. Output: [4, 2, 4, -1, -1].

Practice Questions:

- Next Greater Element I (#496)
- Daily Temperatures (#739)
- Largest Rectangle in Histogram (#84)

Top 'K' Elements

Definition: Uses a Heap (Min-Heap or Max-Heap) to find the K largest or smallest elements in a collection.

Sample Problem: Find the Kth largest element in an array.

Practice Questions:

- Kth Largest Element in an Array (#215)
- Top K Frequent Elements (#347)

Overlapping Intervals

Definition: Involves sorting intervals and then merging or comparing them to find overlaps.

Sample Problem: Merge all overlapping intervals.

Example: intervals = [[1,3],[2,6],[8,10]]. Merged: [[1,6],[8,10]].

Practice Questions:

- Merge Intervals (#56)
- Insert Interval (#57)
- Non-Overlapping Intervals (#435)

Modified Binary Search

Definition: An extension of binary search used to solve problems in rotated sorted arrays or to find specific boundaries.

Sample Problem: Search for a target in a rotated sorted array.

Example: nums = [4, 5, 6, 7, 0, 1, 2], target = 0. Output: 4.

Practice Questions:

- Search in Rotated Sorted Array (#33)
- Find First and Last Position of Element (#34)

Binary Tree Traversal

Definition: Techniques like BFS (Level-order) and DFS (Pre/In/Post-order) to visit all nodes in a tree.

Practice Questions:

- Binary Tree Level Order Traversal (#102)
- Binary Tree Zigzag Level Order Traversal (#103)

Depth First Search (DFS)

Definition: Explores as far as possible along each branch before backtracking. Common in tree and graph problems.

Practice Questions:

- Clone Graph (#133)
- Path Sum II (#113)
- Course Schedule II (#210)

Breadth First Search (BFS)

Definition: Explores all neighbors at the present depth before moving to nodes at the next depth level. Ideal for shortest path in unweighted graphs.

Practice Questions:

- Rotting Oranges (#994)

Word Ladder (#127)

Matrix Traversal

Definition: Treating a 2D matrix as a graph and using DFS or BFS to traverse it.

Practice Questions:

Number of Islands (#200)

Flood Fill (#733)

Backtracking

Definition: A recursive approach to find all (or some) solutions by building candidates and abandoning them ("backtracking") as soon as it determines they cannot lead to a valid solution.

Practice Questions:

Permutations (#46)

Subsets (#78)

N-Queens (#51)

Dynamic Programming (DP)

Definition: Solving complex problems by breaking them into simpler subproblems and storing the results (memoization or tabulation) to avoid redundant work.

Practice Questions:

Climbing Stairs (#70)

House Robber (#198)

Coin Change (#322)